# Systems of measurement

What does it mean to measure something?

Why are we measuring something?

What are we going to do with the measurement? How are we going to use it?

Data can be:

Nominal –name, no order, categories (set of pets)

Ordinal – has an order (like people in a line, tuple)

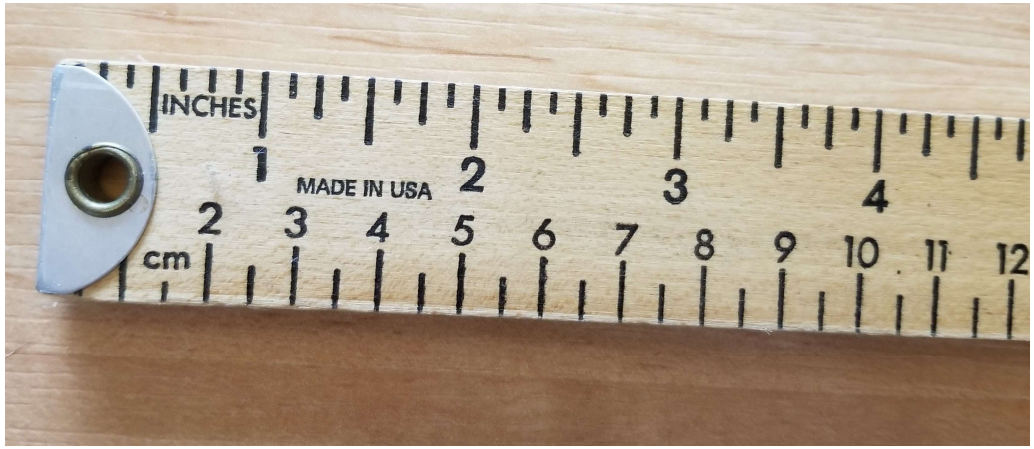
Interval – can add and subtract (like degrees Celsius or Fahrenheit)

Ratio – can multiply and divide (like degrees Kelvin, length)

# There are different types of measurement.

Weight, volume, time, count, and length.

# A type of measurement can have different units.



# What you measure and how you measure is studied in science.

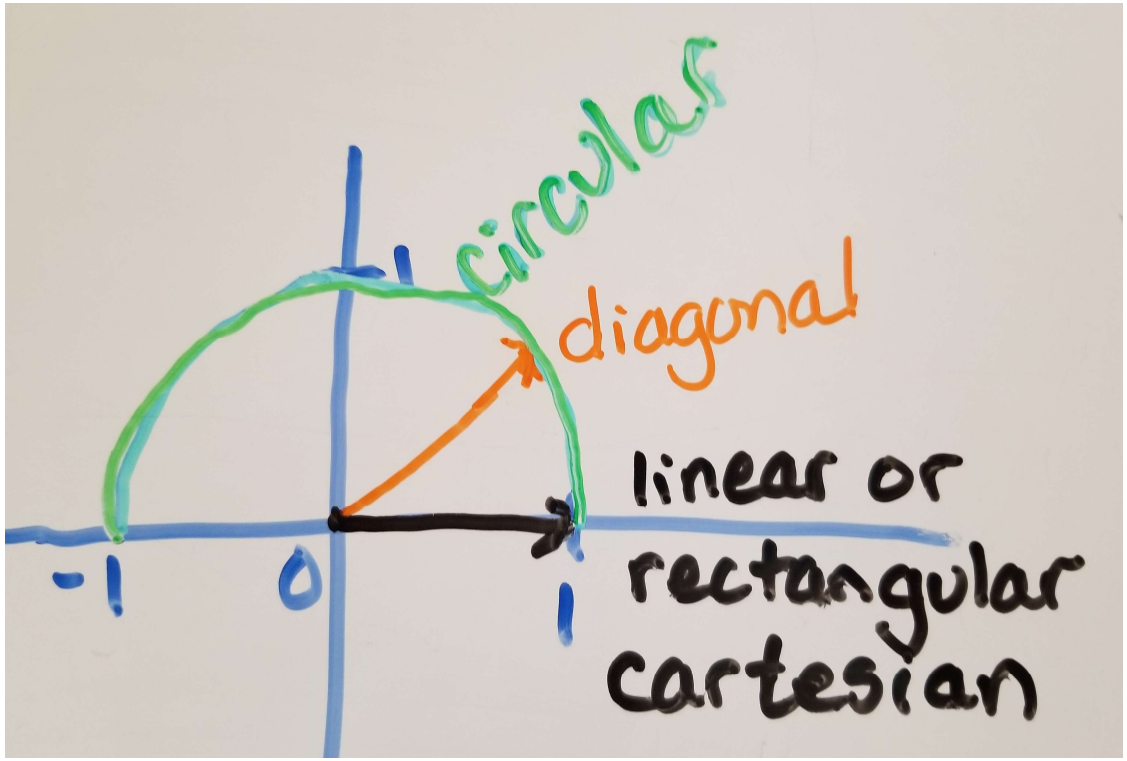
In math, for length, there are three different systems that don't match up for measuring with a measuring tape or string.  This isn't mathematical but how I think about the three different systems.

Rectangular, or cartesian

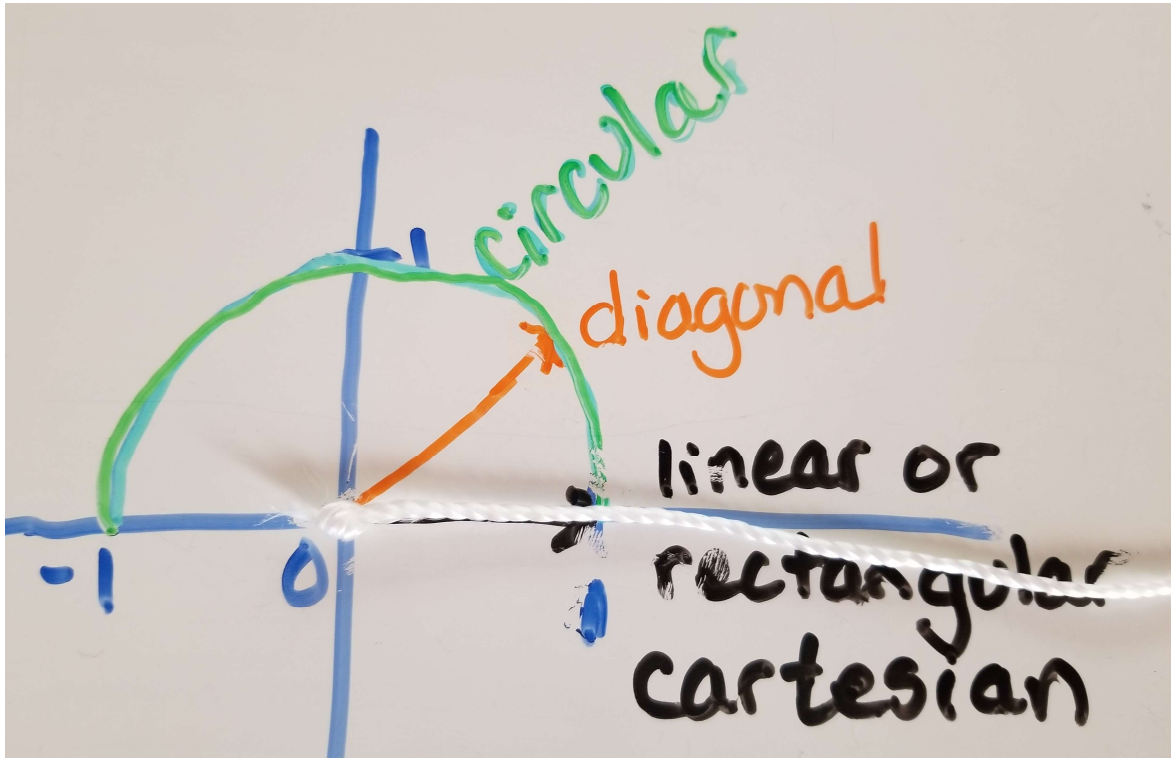
Diagonal

Circular

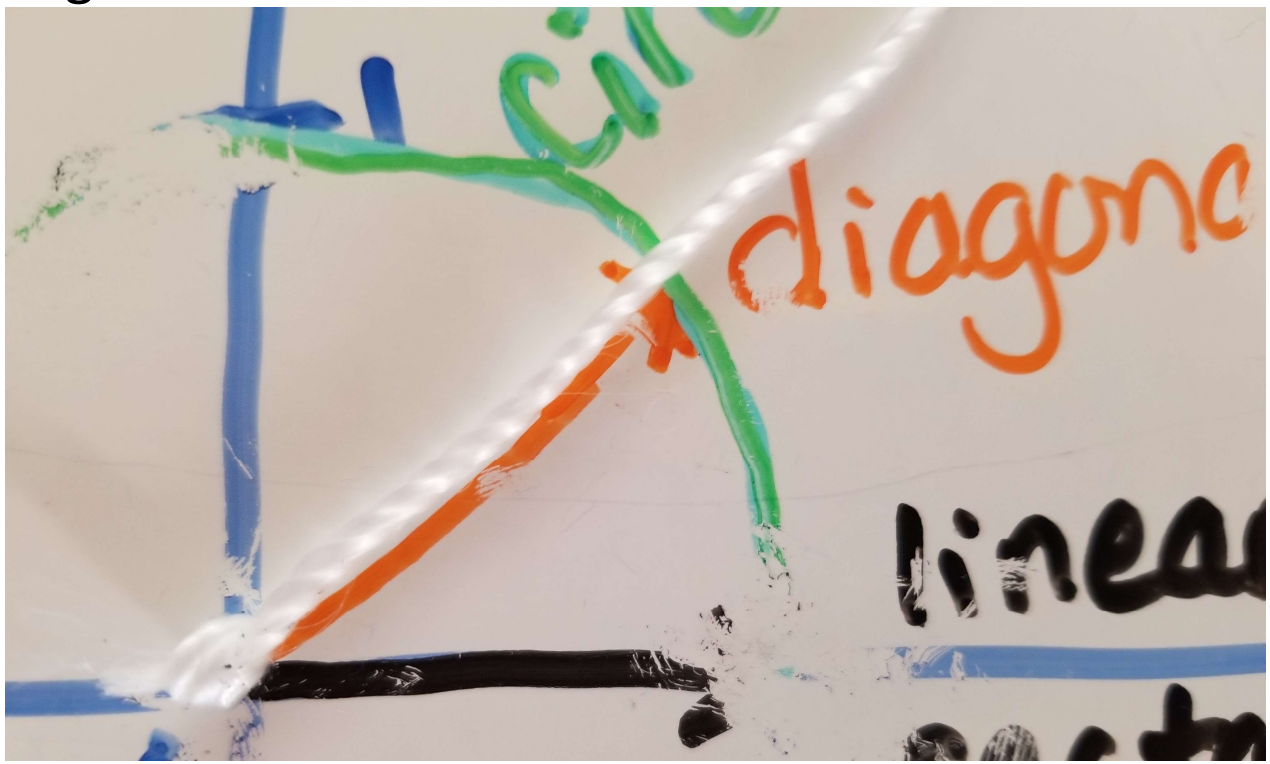
(not official names)



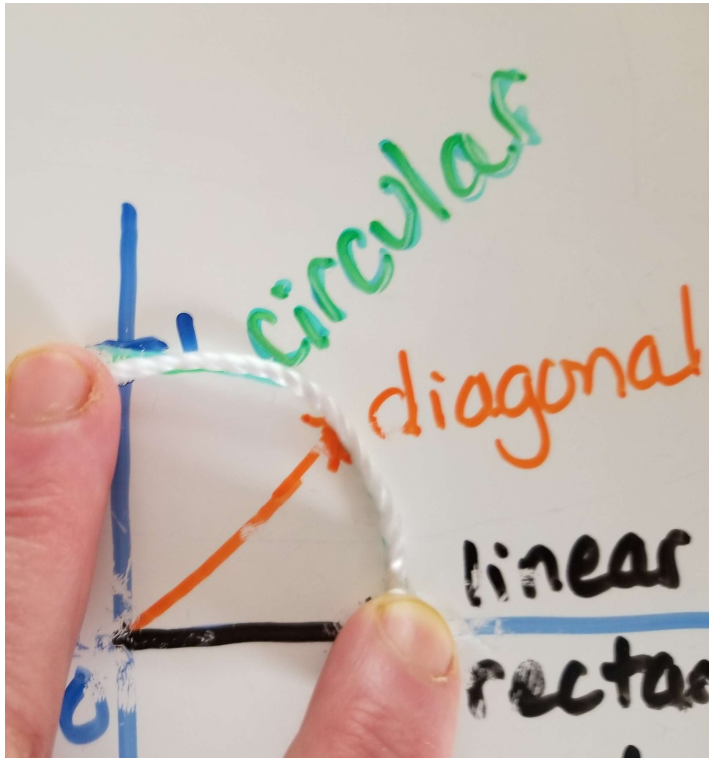
If we take a string and measure along a line on a cartesian system or a measuring tape, with a length of one, we get to one unit on the x and y axes. We are doing math, so we abstract the units.



If we move a length of string with a distance of one on a diagonal, the diagonal still has length 1. But if we try and find the x and y coordinates for that point, we don’t get a rational number. That number can’t be written as a ratio of two integers.



If we try and measure a quarter of a circle, and then see how long that piece of string is on a linear measuring tape, we don’t get a rational number. That number can’t be written as a ratio of two integers.



# Commensurability (mathematics) - Wikipedia

Two numbers are commensurable if their ratio is a rational number. If their ratio is not a rational number, then they are **incommensurable**.

They don’t have a rational measuring system in common.

# Irrational numbers

We have been talking about rational numbers, and we now will be working with irrational numbers.

Irrational numbers cannot be expressed as a ratio of two integers. They give infinitely-long, nonrepeating decimals.

Because we can't write them as a finite decimal number, we give them names.

Square roots of nonperfect squares $\sqrt{}$, pi $\pi$, and Euler's number *e,* are all irrational numbers. $\sqrt{2}, \sqrt{3}, \sqrt{5}$

There are infinitely many rational numbers between every two rational numbers and infinitely many irrational numbers between every two rational numbers on a number line!

Luckily, we only deal with a few irrational numbers:

Square roots of nonperfect squares $\sqrt{}$ (or polynomial roots…)

Euler's number *e*

pi $\pi$